

Stratification and Regrouping for Profile-Coherent Financial Asset Recommendation

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Abstract

Financial asset recommendation (FAR) literature optimises ranking quality and realised return, but no published metric exposes whether recommendations are profile-aligned in the regulatory sense that MiFID II suitability requires. We introduce Profile Coherence at k (PC@ k), scoring the share of a top- k list whose asset risk class lies within ± 1 band of the customer’s declared MiFID II risk band on a four-point ordinal scale. Applying PC@ k to FAR-Trans [Sanz-Cruzado *et al.*, 2024], we find that 18.6% of 228,241 scoreable Buys violate the declared band by two or more bands, with a sharply bimodal per-customer self-discordance distribution. Yet a transaction-level Ordinary Least Squares (OLS) regression on 208,029 priced Buys, with cluster-robust standard errors on `customerID`, finds coherent Buys earn +2.94 percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. We then propose a LightGCN extension with three components: band stratification, a profile-coherent margin loss atop Bayesian Personalised Ranking (BPR), and a regrouping step that reassigns customers to the band implied by their purchases. Across all 69 temporal evaluation splits, stratification with regrouping lifts aggregate PC@10 from 0.794 to 0.949 while also improving nDCG and Recall, and adding the margin loss reaches 0.982, all at negligible aggregate cost to return on investment.

1 Introduction

Financial asset recommendation (FAR) sits at the intersection of personalised ranking and regulatory obligation. The European Markets in Financial Instruments Directive (MiFID II) requires firms to assess suitability before recommending a product: alignment between the product’s risk profile and the customer’s declared risk tolerance, investment horizon, and loss capacity. A recommender that maximises predicted return without conditioning on the customer’s risk band can be an excellent ranker on standard benchmarks but a regulatory liability.

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first open FAR benchmark with both customer profile metadata and a multi-year transaction history: 228,913 Buys over an asset universe of 806 instruments and 29,090 customers (predominantly retail), with MiFID II risk bands per customer and metadata sufficient to derive each asset’s risk class. Of these, 228,241 Buys (99.7%) are *scoreable*: every asset carries a derived risk class, and these are the Buys whose customer also has a declared band, enabling pairwise discordance measurement. The accompanying paper benchmarks LightGCN [He *et al.*, 2020], a graph-based collaborative filter that propagates embeddings over the user-item bipartite graph, Random Forest, a tree-based ensemble that ranks assets from tabular customer and asset features, and others on normalised Discounted Cumulative Gain (nDCG) and Return on Investment (ROI), each evaluated on the top-10 ranked items. We additionally report Recall alongside these as a ranking sanity check. None of those metrics measures profile alignment, and none of the published baselines conditions on the declared risk band.

This paper introduces a profile-alignment lens for FAR and applies it along a diagnostic-to-remedy chain: *diagnose* the prevalence of discordance and *test* whether it carries an economic penalty, *audit* where existing baselines sit on the new axis, and *remedy* the gap with a profile-coherent LightGCN combining stratification, a coherence-margin loss, and customer regrouping. These map onto three research questions:

- **RQ1 (Diagnose and test):** How prevalent is profile-discordance in observed FAR-Trans Buy transactions, and is it a stable customer trait or transaction-level noise? And do profile-discordant Buys earn lower realised 6-month return than coherent ones, once asset volatility, customer segment, and year are controlled for?
- **RQ2 (Audit existing baselines):** At their best published settings, do the chosen FAR-Trans baseline models lift profile coherence above the band-conditional random baseline $\pi(b)$ (the expected coherence of a uniformly random recommender) for every declared band?
- **RQ3 (Remedy):** Can a profile-coherent LightGCN built from band stratification, a coherence-margin loss, and regrouping customers onto their revealed band improve profile coherence over the global baseline, and at what cost to nDCG, Recall, and ROI?

Contributions. We make three contributions, each measured with Profile Coherence at top- k ranked items (PC@ k), a coherence metric with a band-conditional random baseline $\pi(b)$. First, a profile-coherent *margin loss* atop Bayesian Personalised Ranking (BPR) [Rendle *et al.*, 2012] that makes PC@ k a training-time objective. Second, a band *stratification* step that routes each customer to a per-band sub-model. Third, a *regrouping* step that reassigns customers to the band implied by their revealed purchases. An ablation isolates each component.

94 **2 Preliminaries**

This section provides the background needed to read the rest of the paper: the financial asset recommendation problem, the FAR-Trans dataset and benchmark that ground our study, then the two-axis evaluation standard whose blind spot motivates our contribution.

100 2.1 The financial asset recommendation problem

Financial asset recommendation (FAR) identifies and ranks financial securities (stocks, bonds, mutual funds) for an investor according to their suitability. Suitability depends on investor-side signals such as past transactions, declared risk tolerance, investment capacity, and personal goals, and on market-side signals such as asset returns, currency value, and inflation. A FAR system therefore draws on heterogeneous data sources: time-series pricing data, customer profile data, and historical investment transactions.

110 2.2 The FAR-Trans dataset

Most prior FAR models were developed over proprietary or simulated data, which prevents fair cross-method comparison. The only earlier public dataset lacked pricing data and asset identifiers, ruling out price-based approaches. FAR-Trans [Sanz-Cruzado *et al.*, 2024] closes this gap: it is the first public FAR dataset that pairs real asset pricing with real retail-investor transactions, collected from a large European financial institution and spanning January 2018 to November 2022. It provides daily closing prices, Buy and Sell transactions, and customer profile metadata including a declared MiFID II risk band per customer. Together these make both price-based and profile-based recommendation measurable on a single dataset.

2.3 The two-axis benchmark and its blind spot

Alongside the data, FAR-Trans establishes the field’s evaluation standard by benchmarking 11 FAR algorithms, among them LightGCN and Random Forest. The benchmark reports two axes: normalised Discounted Cumulative Gain (nDCG@10) for next-purchase ranking quality and Return on Investment (ROI@10) for realised profitability, each over the top-10 ranked items. A subsequent study [Sanz-Cruzado *et al.*, 2026] shows that these two axes are not aligned on FAR-Trans: transaction-based and profitability-based metrics are negatively correlated, so a model that wins on one tends to lose on the other, and the authors conclude that ranking quality and realised return capture genuinely different objectives. Crucially, neither axis measures whether a recommendation

aligns with the customer's declared MiFID II risk band. That
regulatory suitability dimension is the blind spot this paper
targets.

3 Related Work

Building on the FAR-Trans benchmark and its two-axis evaluation standard introduced in Section 2, we position our contribution against three strands of prior work: profile-aware FAR architectures, risk-aware re-ranking, and fairness-aware recommendation.

3.1 Profile-aware FAR architectures

Ghiye *et al.* [2024] extend LightGCN with a causal graph con- 148
 volution and a sliding-window training scheme for credit-bond 149
 recommendation, capturing temporal drift in user interests 150
 via embedding updates restricted to recent interactions. Their 151
 result supports the broader point that LightGCN can be mod- 152
 ified to carry domain-specific structure beyond plain Bayesian 153
 Personalised Ranking (BPR), and motivates the choice of 154
 LightGCN as the backbone for our extension. 155

3.2 Risk-aware FAR via post-hoc re-ranking

Sakurai *et al.* [2026] propose a risk-aware utility re-ranking (RURA) procedure on FAR-Trans: a mean-variance re-ranking objective conditioned on the customer’s MiFID II risk tolerance is applied to a base ranker’s top list, lifting ROI@10 well above the base ranker while keeping nDCG@10 only on par with other risk-aware re-rankers, far below the base ranker’s own nDCG@10. RURA optimises a risk-tolerance utility but does not quantify how aligned the resulting list is with the declared band in terms of ordinal distance. PC@ k is complementary: it is a metric for that alignment, and is computable on any ranker (with or without re-ranking).

3.3 Fairness-aware recommendation

Fairness-constrained recommender systems enforce group-level quality guarantees such as equalised performance across demographic groups or item providers [Zhao *et al.*, 2024]. PC@ k can be viewed through a similar lens: it measures per-group recommendation quality where groups are defined by declared MiFID II risk bands, and PC-lift@ k adjusts for how easy each band is to satisfy by chance, so bands can be compared on equal footing. The key difference is that PC@ k targets a regulatory suitability constraint rather than a social fairness objective.

All in all, existing FAR work optimises profitability or ranking, with risk-tolerance entering only as a post-hoc utility weight. We bring suitability inside the loop: $PC@k$ scores coherence on the same top- k lists the benchmarks already use, and a profile-coherent objective optimises for it during training instead of filtering for it afterwards.

4 RQ1: Statistical Analysis of Profile Coherence

RQ1 establishes the diagnostic axis of the paper. It asks how prevalent profile-discordance is in observed FAR-Trans Buy transactions, whether it is a stable customer trait or transaction-level noise, and whether it carries an economic penalty once

asset volatility, customer segment, and year are controlled for. We first define the risk bands and the asset risk-band assignment the entire paper relies on, formalise the Profile Coherence metric, describe the dataset, then present the diagnostic analysis and the return-premium regression in turn.

4.1 Risk Bands

Ordinal band scale

Customers and assets share the same 4-point ordinal scale running from Conservative to Income to Balanced to Aggressive, encoded 0 to 3. The customer band b_u is taken from FAR-Trans’s `riskLevel` field. FAR-Trans provides both declared values and regression-imputed `Predicted_*` entries, and both are mapped to the same ordinal scale. Customers whose entry is `Not_Available` or missing are excluded from all coherence computations. The asset band b_i is assigned by the three-step hierarchical rule below, applied in order.

Step 1: subcategory metadata

Mutual funds are labelled directly from their subcategory: Money Market funds are treated as Conservative, Bond funds as Income, Balanced as Balanced, and Equity or Large Cap as Aggressive. Bond instruments are labelled by issuer type: Government as Conservative, Corporate as Income, and any remaining bond subcategory as Income by default.

Step 2: volatility quartiles for stocks

Stocks carry no subcategory tag we can use, so they fall through to a volatility rule. For each ISIN we compute the annualised standard deviation of daily log returns on a trailing 252-day window, then bucket all stocks into the four bands by the stock-only quartile thresholds (q_1, q_2, q_3) .

Step 3: residual default

Any asset that falls through both prior steps is assigned to Balanced.

Applying the three steps above produces an asset-band distribution of (190, 333, 105, 178) across (Conservative, Income, Balanced, Aggressive) over $|A| = 806$ assets, which feeds the band-conditional random baseline $\pi(b)$.

4.2 Profile Coherence

Pairwise discordance

$$d(u, i) = |b_u - b_i| \in \{0, 1, 2, 3\}. \quad (1)$$

A recommendation is *profile-coherent* iff $d \leq 1$. We adopt this ± 1 band tolerance by design, to mirror MiFID II’s adjacency allowance for borderline suitability rather than to penalise every single-band drift as a violation.

Aggregation

$$\text{PC}@k(u) = \frac{1}{k} |\{i \in \text{top}_k(u) : d(u, i) \leq 1\}|. \quad (2)$$

Two implementation rules complete the definition. First, recommenders returning fewer than k items use the actual number of items returned as the denominator. Second, customers with no assigned risk band ($b_u = \emptyset$) are dropped from the aggregate. Per-split $\text{PC}@k$ is the unweighted mean across eligible customers and the headline numerics average over all splits.

Band-conditional random baseline

The closed-form expectation of $\text{PC}@k$ under a uniformly random recommender for band b is given by

$$\pi(b) = \frac{|\{i \in A : |b - b_i| \leq 1\}|}{|A|}, \quad (3)$$

where A is the asset universe. With the asset-band distribution from Section 4.1, this yields $\pi(b) = (0.65, 0.78, 0.76, 0.35)$.

PC-lift@k

Per-user $\text{PC}@k$ is normalised against the random reference at the customer’s own band:

$$\text{PC-lift}@k(u) = \frac{\text{PC}@k(u)}{\pi(b_u)}. \quad (4)$$

Lift > 1 beats the band-conditional random rate, while lift $= 1$ matches it. Because 77.9% of FAR-Trans assets sit in the Conservative, Income, and Balanced bands, a recommender that pushes Income assets at everyone would score well on raw $\text{PC}@k$ for any user whose tolerance set includes Income. Dividing by $\pi(b_u)$ removes this advantage, so bands that are easier to satisfy by chance are not credited for it and the lift values can be compared across bands.

4.3 Dataset

In addition to what was introduced in Section 1, FAR-Trans [Sanz-Cruzado *et al.*, 2024] provides 159,135 Sell transactions. Of the 29,090 customers, 28,770 (98.9%) carry a declared or regression-imputed MiFID II risk band, and the remaining 320 are excluded from all coherence computations. Customer profiles also include `customerType` and `investmentCapacity`, both of which enter our regression study later as controls.

4.4 Prevalence and Structure of Discordance

We characterise discordance in four slices, starting from the pooled transaction rate and drilling into the customer, band, and year dimensions in turn. At the transaction level (Figure 1), of 228,241 scoreable Buys:

1. 18.6% violate the declared band by two or more steps.
2. 81.4% are coherent, of which 34.1% are strictly coherent ($d = 0$).
3. Mean discordance is 0.874 bands across all scoreable Buys, indicating that when mismatches occur they are usually around a single band step.

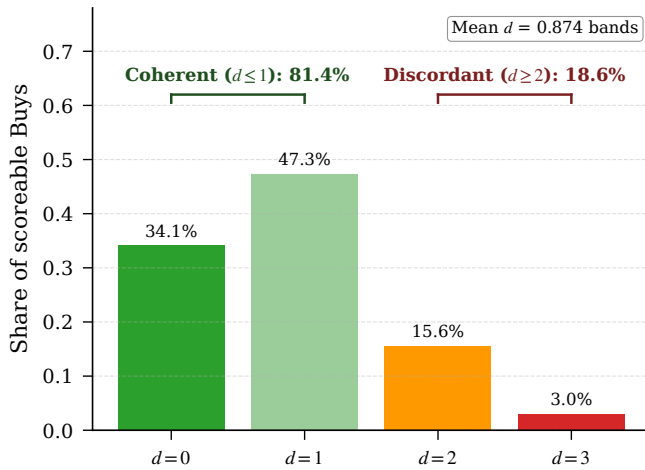


Figure 1: Distribution of pairwise discordance d over 228,241 scoreable Buys. Most mass sits at $d \leq 1$ (coherent), with non-negligible density at $d \geq 2$ (discordant).

The pooled rate cannot tell us whether these violations are i.i.d. transaction-level slips spread thinly across the customer base or a structural trait concentrated in a subset. Slicing by customer settles the question. The per-customer discordant share (Figure 2) is sharply bimodal, concentrating in the lowest and highest bins rather than clustering near the 18.6% population mean. More explicitly, 64.4% of banded customers are fully coherent across every Buy and 17.2% are fully discordant.

Hartigan's dip test on the 28,770 per-customer values rejects unimodality ($D = 0.086$, $p < 10^{-15}$), confirming that discordance is a persistent customer-level trait rather than i.i.d. transaction-level noise.

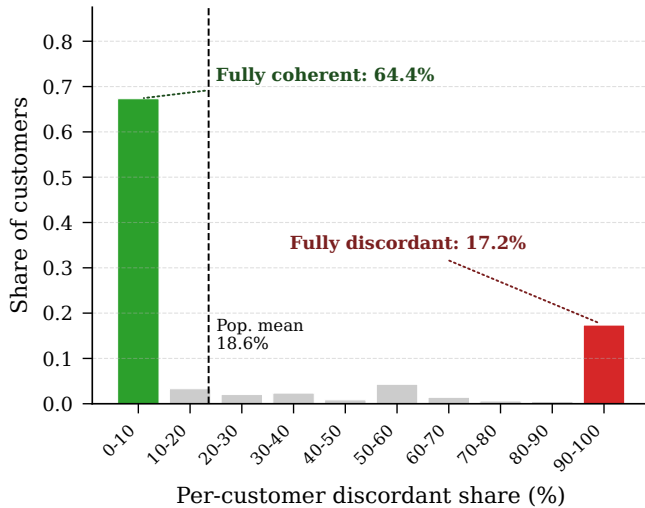


Figure 2: Per-customer share of discordant Buys is sharply bimodal, concentrating in the lowest and highest bins and ruling out i.i.d. transaction-level noise.

Since the trait is customer-level, the next question is which customers carry it. Coherence per declared band is U-shaped

(Figure 3). The two centre bands sit around 90%, while Conservative drops to 45.1% (possibly reflecting a “reach for yield” tendency) and Aggressive to 56.2% (possibly reflecting a regression toward safer assets). This concentration of mismatch at the ordinal extremes corroborates the bimodality finding from the customer slice.

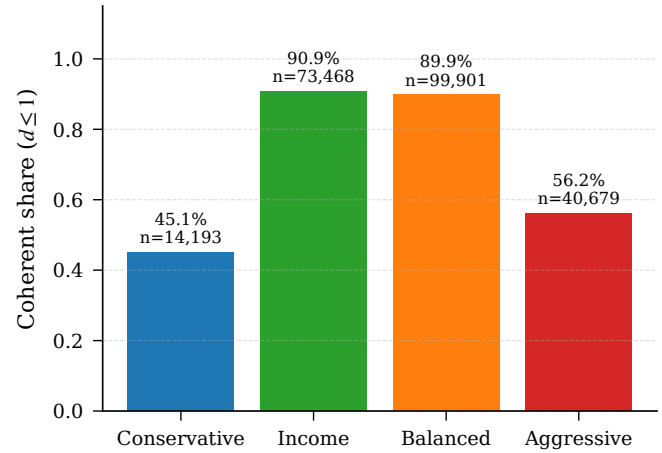


Figure 3: Coherence share by declared band. The U-shape supports the bimodality result of Figure 2: discordance concentrates on the ordinal extremes.

A final slice rules out a regime-driven artefact. Mean discordance is flat across 2019 to 2022 (within 0.02 bands, see Figure 4), so the pattern is not the product of any single year's market conditions. The 2018 value (0.98) sits higher but covers a small slice. MiFID II came into force on 2018-01-03, the exact start of FAR-Trans, so we hypothesise the elevation as a transition effect. However, this hypothesis is not tested and is potential future work.

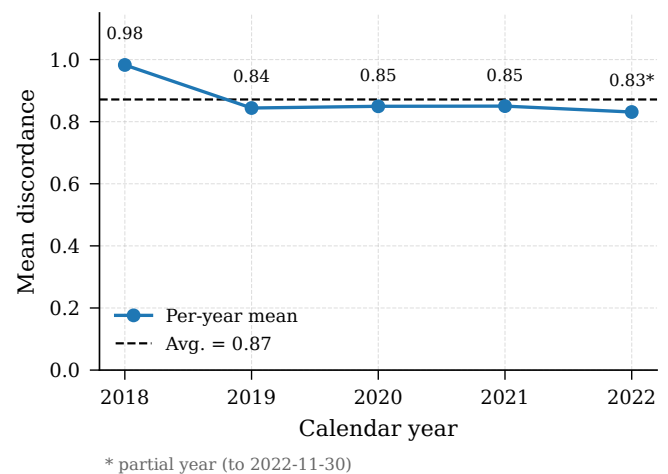


Figure 4: Mean discordance by calendar year. The 2019 to 2022 cluster sits within 0.02 bands, with 2018 elevated (transition-effect hypothesis).

4.5 Conditional Return Premium

We have shown that discordance is a persistent customer-level trait concentrated at the ordinal extremes. The next question is whether it carries an economic cost: do discordant Buys earn lower returns than coherent ones?

Regression specification

We test whether profile-coherent Buys earn different realised returns from discordant ones via a transaction-level Ordinary Least Squares (OLS) regression on 208,029 priced Buys with cluster-robust standard errors on `customerID` and a two-sided Wald test at $\alpha = 0.05$. Clustering is warranted because each customer contributes roughly 8 Buys on average, so within-customer residuals share unobserved shocks. The specification is:

$$\begin{aligned} r = & \beta_0 + \beta_1 \cdot \text{is_coherent} \\ & + \beta_2 \cdot \text{asset_volatility} \\ & + \mathbf{C}(\text{type})\gamma \\ & + \mathbf{C}(\text{year})\delta \\ & + \varepsilon, \end{aligned} \quad (5)$$

where $r = (p_{\text{end}} - p_{\text{start}})/p_{\text{start}}$ is the 6-month realised return as a decimal fraction (e.g., $r = 0.03$ means a 3 percentage-point gain) and `is_coherent` is binary ($d \leq 1$).

Prices are matched via a backward-nearest join: the start price is the most recent close on or before the trade date, and the end price is the most recent close on or before the date six months later. Transactions without a price match within 7 days at either endpoint are dropped. The coefficient of interest is β_1 on `is_coherent`, and the null hypothesis $H_0: \beta_1 = 0$ states that coherence has no conditional effect on 6-month returns.

Results

Table 1 reports the full coefficient table. The key results on the coherence premium are:

- **Conditional premium.** Coherent Buys earn +2.94 percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year.
- **Significance.** The positive sign with $p = 2.3 \times 10^{-13}$ provides strong evidence against the prior that regulatory coherence imposes a return tax.

That said, this premium is an observational association rather than a causal effect, a caveat we return to in Section 4.6.

The adjusted R^2 is 0.051: the covariates explain little of the variation in individual returns, as expected when idiosyncratic price movements dominate. This concerns prediction and does not affect our hypothesis test. The coherence premium is an average difference in return between coherent and discordant Buys, and even with a low R^2 it is estimated precisely. We cluster the standard errors by `customerID` so that repeated Buys from the same customer are not counted as independent evidence.

The volatility coefficient is -10.25 pp per unit, consistent with low-volatility outperformance over 2019 to 2022 and the 2022 rate shock that hit high-beta names. All four year

dummies reject H_0 , reflecting distinct market regimes relative to the 2018 reference year:

1. 2019 coefficient captures the equity recovery from the late-2018 selloff.
2. 2020 is the strongest at +21.07 pp driven by the post-COVID V-shaped rebound.
3. 2021 reflects continued recovery under accommodative monetary policy before late-2021 rate-hike expectations set in.
4. 2022 coefficient turns negative at -1.95 pp as aggressive Fed hikes triggered a broad fixed-income drawdown.

All four `customerType` dummies fail to reject H_0 , indicating that segmentation contributes no detectable return signal. Together with the bimodality found above, this suggests that customer type does not capture the persistent discordance trait either.

Taken together, these controls barely move the premium: the raw slice gap is +3.31 pp (unconditional mean 6-month return: 4.23% coherent vs 0.92% discordant), whereas the `is_coherent` coefficient, which already conditions on volatility, segment, and year, is +2.94 pp. Adding the controls therefore absorbs only $3.31 - 2.94 = 0.37$ pp ($\sim 11\%$ of the unconditional gap), so coherence is not a volatility, segment, or year confound. The coherence premium therefore survives all controls we can measure, motivating the question of whether existing recommenders produce coherent lists at all.

4.6 Limitations

Two caveats bound the RQ1 findings.

- *Observational nature.* The coherence premium is associational as unobserved confounders such as financial literacy, advisor channel, account constraints, and prior portfolio composition are not ruled out by the volatility, segment, or year controls. Similarly, the elevated 2018 mean discordance is hypothesised as a MiFID II transition effect but not directly tested against a pre-MiFID II counterfactual.
- *Asset-band assignment.* The asset bands b_i are not provided by FAR-Trans but assigned by the hierarchical heuristic in Section 4.1 rather than an externally validated classification, so any systematic mislabelling, especially from the volatility-quartile step for stocks, propagates into all coherence numbers.

5 RQ2: Coherence Weaknesses of FAR-Trans Baselines

With a confirmed return premium for coherent Buys from our regression analysis (Section 4), we now audit whether the two FAR-Trans baselines actually deliver profile-coherent recommendations across all four risk bands. We choose these two because, in the original FAR-Trans benchmark of 11 algorithms, LightGCN is the strongest ranking model (highest nDCG@10) and Random Forest the strongest profitability model (highest ROI@10), with the authors noting that each leads only on its own objective.

Term	Estimate	Std. Error	95% CI	<i>p</i>
Intercept	-0.0471	0.0141	[-0.0746, -0.0195]	8.1×10^{-4}
is_coherent	+0.0294	0.0040	[+0.0215, +0.0372]	2.3×10^{-13}
asset_volatility	-0.1025	0.0165	[-0.1349, -0.0701]	5.5×10^{-10}
C(customer_type) [Legal Entity]	+0.0154	0.0486	[-0.0798, +0.1106]	0.751
C(customer_type) [Mass]	+0.0011	0.0132	[-0.0248, +0.0269]	0.936
C(customer_type) [Premium]	+0.0058	0.0135	[-0.0206, +0.0322]	0.667
C(customer_type) [Professional]	+0.0247	0.0169	[-0.0084, +0.0578]	0.143
C(year) [2019]	+0.1759	0.0129	[+0.1507, +0.2012]	1.4×10^{-42}
C(year) [2020]	+0.2107	0.0061	[+0.1986, +0.2227]	6.2×10^{-259}
C(year) [2021]	+0.0754	0.0053	[+0.0651, +0.0858]	4.1×10^{-46}
C(year) [2022]	-0.0195	0.0045	[-0.0283, -0.0107]	1.4×10^{-5}

Table 1: Full RQ1 OLS coefficient table on 208,029 priced Buys, cluster-robust SE on customerID. Reference cell: Inactive customerType (the alphabetically first segment in FAR-Trans), year 2018, asset_volatility = 0, is_coherent = 0.

RQ2 asks: at their best published settings, do the chosen FAR-Trans baseline models lift profile coherence above the band-conditional random baseline $\pi(b)$ for every declared band? We first describe the temporal evaluation protocol and the baseline grids, then present the per-band audit.

5.1 Temporal Split Protocol

Evaluation uses 69 temporal splits with approximately bi-weekly split points on an evenly spaced trading-day grid running from August 2019 to December 2021, with per-split test windows extending evaluation coverage to May 2022. The two-week split cadence follows the original FAR-Trans benchmark [Sanz-Cruzado *et al.*, 2024].

For each split, the training set contains all Buys in $[0, \text{time_point}]$ and the test set contains all Buys in $[\text{time_point}, \text{test_end}]$, deduplicated against training and filtered to the training-asset universe. Eligible customers have at least one interaction in both training and test, while eligible assets have prices on both endpoints.

5.2 Baselines and Hyperparameter Grids

Table 2 reports the search grids, fixed values, and best-trial selections for the two FAR-Trans baselines. Both grids are swept with Ray Tune, with LightGCN tuned on average nDCG@10 and Random Forest on average ROI@10 across the 69 splits.

5.3 Per-Band Audit Method

For each (declared band, model) cell, we average per-customer PC@10 unweighted over all (customer, split) pairs. Per-band lift is then the ratio of that cell mean to the random baseline $\pi(b)$, which adjusts for how easy each band is to satisfy by chance, as defined in Section 4.2.

5.4 Results

We begin by comparing the aggregate performance of the two baselines at their primary-metric optima, summarised in Table 3:

- **LightGCN** wins on ranking quality (nDCG@10 of 0.330, Recall@10 of 0.495, PC@10 of 0.794) but loses on ROI@10 at -0.5%/mo.

Parameter	Search space	Best	Role
LightGCN (tuned on average nDCG@10)			
embedding_dim	{64, 128}	64	grid
num_layers	{2, 3}	3	grid
learning_rate	$\{10^{-2}, 10^{-3}\}$	10^{-3}	grid
weight_decay	10^{-5}	n/a	fixed
keep_prob	0.6	n/a	fixed
num_epochs	50	n/a	fixed
batch_size	1024	n/a	fixed
Random Forest (tuned on average ROI@10)			
num_estimators	{20, 30, 40, 50}	40	grid
max_depth	{15, 25, 50}	50	grid
horizon	6 months	n/a	fixed
random_state	42	n/a	fixed

Table 2: Baseline hyperparameter grids and best-trial values. “grid” rows are searched, “fixed” rows are held constant.

- **Random Forest** wins on ROI@10 at +1.4%/mo but ranks near random (nDCG@10 of 0.019, Recall@10 of 0.036).
- **Aggregate PC@10** looks adequate for both (LightGCN $1.17\times$ lift, Random Forest $1.06\times$).

However, aggregate PC@10 averages across customer bands whose random baselines $\pi(b)$ vary from 0.35 to 0.78, so a single number can mask which bands are actually being served. Table 4 decomposes PC@10 by declared band to expose these differences:

- **Lift by band.** Across Conservative, Income, Balanced, and Aggressive, per-band lift rises from 0.75 to 1.13, 1.17, and 1.35 for LightGCN, and from 0.52 to 0.70, 1.03, and 1.88 for Random Forest.
- **Monotonic skew.** For both, lift rises monotonically with the declared band: recommendations skew towards higher-risk assets.
- **Flatter LightGCN.** LightGCN’s range is flatter, which we attribute to its per-user embeddings shifting each customer’s top-*k* partly toward their tolerance window, whereas Random Forest carries no customer signal. Aggressive’s high lift is partly mechanical:

Model	nDCG@10	ROI@10	Recall@10	PC@10	PC-lift@10
LightGCN	0.330	-0.005	0.495	0.794	1.17×
RF	0.019	+0.014	0.036	0.667	1.06×

Table 3: RQ2 aggregate metrics at primary-metric optima, averaged over 69 splits. ROI@10 is per-month. PC-lift@10 is PC@10 divided by the band-conditional random baseline $\pi(b)$.

$\pi(\text{Aggressive}) = 0.35$ is the lowest band, so any model skewing higher-volatility clears that bar easily. Neither baseline has band labels or a band-aware loss, so per-band coherence is a passive byproduct of training rather than a controlled output.

Model	Band	N	$\pi(b)$	PC@10	PC-lift@10
RF	Conservative	5,348	0.65	0.34	0.52
RF	Income	45,142	0.78	0.54	0.70
RF	Balanced	61,181	0.76	0.79	1.03
RF	Aggressive	25,367	0.35	0.66	1.88
LightGCN	Conservative	5,348	0.65	0.48	0.75
LightGCN	Income	45,142	0.78	0.88	1.13
LightGCN	Balanced	61,181	0.76	0.89	1.17
LightGCN	Aggressive	25,367	0.35	0.47	1.35

Table 4: Per-band PC@10 and PC-lift@10 versus the band-conditional random baseline $\pi(b)$ for the two FAR-Trans baselines at their primary-metric optima, aggregated over 69 splits. PC-lift@10 = PC@10 / $\pi(b)$.

5.5 Limitations

First, the audit evaluates each baseline only at its primary-metric optimum (nDCG@10 for LightGCN, ROI@10 for Random Forest), so the per-band picture is conditioned on those operating points rather than swept across the full grid.

Second, per-band lift should be read alongside its $\pi(b)$ reference rather than in isolation, since low- $\pi(b)$ bands such as Aggressive clear the bar more easily.

6 RQ3: Stratified Profile-Coherent LightGCN

RQ2 showed that both baselines under-serve at least one risk band, with the deficit concentrated on the Conservative segment for LightGCN and on both Conservative and Income for Random Forest. RQ3 asks whether a band-aware architecture can close this gap without sacrificing ranking quality or return. We propose a profile-coherent LightGCN (PC-LightGCN) extension built from three components:

- **Stratification.** A band-stratified architecture that routes each customer to a per-band sub-model.
- **Coherence margin.** A coherence margin added to the BPR objective.
- **Regrouping.** A step that reassigns customers to the band implied by their purchases.

We describe each component and the ablation setup, then evaluate the resulting configurations against the baseline LightGCN.

6.1 Backbone

We choose LightGCN as the backbone because its differentiable loss and per-customer embeddings allow us to attach a coherence margin directly to the BPR objective. Random Forest is not extended because it ranks assets by predicted return from technical indicators alone and carries no customer-level signal to condition on.

6.2 Stratified architecture

We train four band-stratified sub-models, one per declared MiFID II risk band (Figure 5). Each sub-model’s training graph contains only the interactions of customers assigned to that band. At inference, each customer is routed to the sub-model matching their declared band. Customers with no declared band default to the Balanced sub-model.

6.3 Profile-coherent margin loss

BPR trains the model to score an asset the customer actually bought (the positive item) higher than a randomly sampled unobserved asset (the negative item). Independently of this BPR sampling, we add a second margin term that samples one coherent asset i_c (satisfying $d(u, i_c) \leq 1$) and one discordant asset i_d (satisfying $d(u, i_d) > 1$) from the sub-model’s asset universe, and pushes the coherent score above the discordant score:

$$\mathcal{L}_{\text{coh}}(u) = \text{softplus}(s(u, i_d) - s(u, i_c)), \quad (6)$$

where $s(u, i)$ is the LightGCN inner-product score between customer u and asset i . Here $\text{softplus}(x) = \ln(1 + e^x)$ is a smooth approximation of the hinge $\max(0, x)$: the penalty is large when the discordant asset outscores the coherent one and decays smoothly toward zero as the coherent score pulls ahead. We use softplus rather than a hard hinge because it is differentiable everywhere, giving stable gradients throughout training, and it matches the form of the standard BPR loss $-\ln \sigma(\cdot)$, keeping the two objectives consistent.

6.4 Total loss

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \omega \|\Theta\|_2^2 + \lambda \cdot \mathcal{L}_{\text{coh}}, \quad (7)$$

Here ω is the L2 weight decay on embedding parameters Θ and $\lambda \geq 0$ controls how strongly the coherence term influences training. BPR remains the primary ranking signal, and only the treatment trial activates the coherence term.

6.5 Customer regrouping

The stratification and margin-loss components both condition on the customer’s declared MiFID II band. RQ1 showed that this declared band is frequently inconsistent with revealed behaviour, especially at the ordinal extremes. The third component therefore *regroups* customers: each banded customer

is reassigned to the band implied by their revealed purchase behaviour, taken as the lower median of the risk classes of the assets they actually bought (even-count ties resolve to the more conservative band). The stratified, profile-coherent model is then retrained and routed on the regrouped bands.

6.6 Ablation and evaluation setup

The stratified PC-LightGCN inherits the best LightGCN trial’s backbone hyperparameters (Table 2). Two trials run on the same 69 splits: $\lambda = 0$ trains stratified BPR alone, isolating the architecture, and $\lambda = 1$ adds the coherence margin on top.

This separates the gains from stratification and from the PC-loss, with every metric reported against the baseline LightGCN. Regrouping is evaluated by rerunning the same $\lambda \in \{0, 1\}$ ablation on the regrouped bands.

6.7 Results

We read the declared-band ablation off each table and figure first, comparing the baseline against the two declared trials ($\lambda \in \{0, 1\}$). The regrouped rows and columns follow in Section 6.8. In Table 5, the full treatment’s coherence gain decomposes into a cheap stratification step and a costlier coherence-margin step.

- **Full treatment** ($\lambda = 1$). Lifts PC@10 from 0.794 to 0.918, a +12.4 pp gain, and PC-lift from $1.17\times$ to $1.40\times$.
- **Stratification alone** ($\lambda = 0$). Supplies +2.7 pp of PC@10, about a quarter of the total, at virtually no cost elsewhere: nDCG@10 +0.2 pp, Recall@10 +0.6 pp, and ROI@10 -0.2 pp/mo, none of which registers as a meaningful effect.
- **Coherence margin** ($\lambda = 1$). Buys the remaining +9.7 pp of PC@10, roughly four-fifths of the total, at a bounded ranking cost of -2.3 pp nDCG@10 and -3.4 pp Recall@10.
- **ROI@10**. Essentially flat across all three trials: -0.2 pp/mo for stratification, 0.0 pp/mo for the margin step alone, and -0.2 pp/mo for the full treatment. The small cost lives in the stratification step, not the margin, and is too small to distinguish from zero.

Together with the RQ1 coherence premium, this null ROI effect shows the recommender can be made more profile-coherent without giving up the return story. The win rate, the fraction of splits on which a configuration beats the baseline, captures how consistently a gain is paid rather than resting on a single strong split (Figure 6):

- **Coherence wins everywhere**. The full treatment posts a 100% win rate on PC@10 across all 69 splits.
- **Cost is uniform too**. The full treatment’s nDCG@10 and Recall@10 fall below baseline on all but a handful of splits (a 7% win rate), so the ranking penalty is a consistent shift rather than a few bad splits.

The paired trial-minus-baseline deltas (Figure 7) make each metric’s shift visible split by split.

- **PC@10 always positive**. PC@10 is positive on every one of the 69 splits for all three configurations, so the aggregate coherence gains are not an averaging artefact.

- **Ranking deltas match the table**. They track the aggregate costs from Table 5: consistently negative for the declared margin trial and hugging zero for stratification alone.

Lastly, Table 6 decomposes PC-lift@10 by band, tracing how the intervention resolves the per-band deficit from RQ2 (declared columns):

- **Conservative**. Sits at $0.75\times$ under the baseline, climbs to $0.91\times$ under stratification alone but stays below the random baseline, and only crosses it at $1.19\times$ once the coherence margin is added.
- **Aggressive**. Non-monotonic: stratification alone reduces lift from $1.35\times$ to $1.17\times$, likely because the smaller Aggressive training graph yields lower-quality embeddings in isolation, but the coherence margin more than recovers the loss at $1.97\times$.
- **Income and Balanced**. Already above-random under the baseline, they improve modestly without being sacrificed.

Under declared $\lambda = 1$, all four bands clear the random baseline for the first time. Reading the three views together is what motivates regrouping. The coherence margin spends ranking quality to buy coherence: Figure 6 shows nDCG@10 and Recall@10 dropping on almost every split while ROI@10 is barely touched, and Table 5 returns only a modest aggregate lift in return. Table 6 shows where that cost goes: the margin is largely spent rescuing a single band, dragging Conservative from $0.91\times$ past random to $1.19\times$. Penalising incoherence during training is therefore an expensive patch for what RQ1 already flagged as noisy declared bands.

The natural alternative is to correct the labels at the source instead of fighting them with a penalty: regroup each customer onto their revealed band first, which should lift coherence under the hood.

6.8 Effect of Regrouping

Rerunning the ablation on the regrouped bands changes the picture in three ways:

- **Strongest and free lever**. Regrouping without the PC-loss (regrouped $\lambda = 0$) lifts PC@10 to 0.949, a +15.5 pp gain that already exceeds the declared full treatment (0.918), and posts the highest nDCG@10 (0.352, +2.2 pp) and Recall@10 (0.522, +2.7 pp) of all five configurations with ROI@10 flat, each winning on every split. Where the margin buys coherence by spending ranking quality, regrouping raises both together.
- **Dominates declared stratification**. Against declared $\lambda = 0$, it adds +12.8 pp PC@10, +2.0 pp nDCG@10, and +2.1 pp Recall@10 on every split and removes the small ROI cost. Since nDCG and Recall are band-independent, this is direct evidence that the declared bands were injecting label noise.
- **Margin adds little once corrected**. The margin’s marginal PC@10 gain collapses from +9.7 pp on the declared bands to +3.3 pp on the regrouped bands (regrouped $\lambda = 1$ reaches 0.982), still costing -1.9 pp

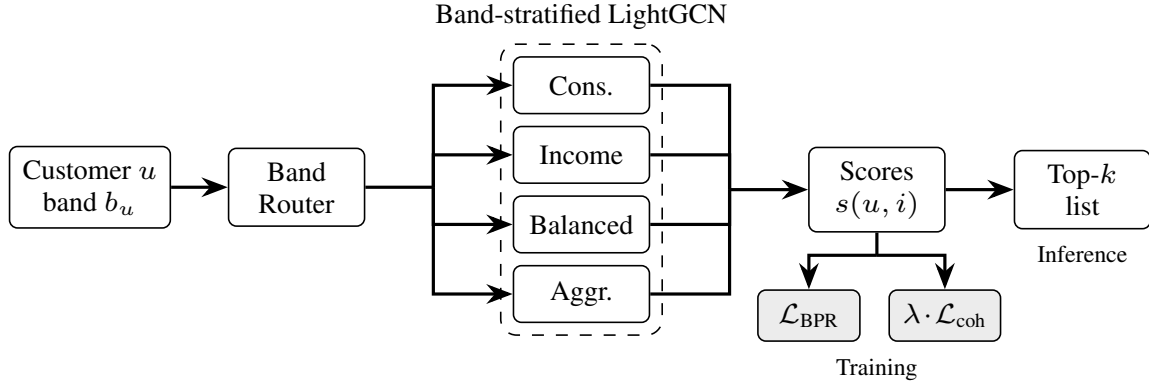


Figure 5: PC-LightGCN architecture. Each customer is routed by declared MiFID II band to a dedicated sub-model. Training combines $\mathcal{L}_{\text{BPR}}$ with a coherence margin $\lambda \cdot \mathcal{L}_{\text{coh}}$. At inference the sub-model produces the top- k list directly.

Trial	nDCG@10	ROI@10	Recall@10	PC@10	PC-lift@10
Baseline	0.330	-0.005	0.495	0.794	1.17×
Declared bands					
Stratify ($\lambda=0$)	0.332	-0.007	0.501	0.821	1.19×
Stratify + PC-loss ($\lambda=1$)	0.309	-0.007	0.467	0.918	1.40×
Regrouped bands					
Stratify ($\lambda=0$)	0.352	-0.006	0.522	0.949	n/a
Stratify + PC-loss ($\lambda=1$)	0.333	-0.005	0.493	0.982	n/a

Table 5: RQ3 aggregate metrics, averaged over 69 splits. ROI@10 is per-month. PC-lift@10 = $\text{PC@10} / \pi(b_u)$ and is n/a for the regrouped trials, whose shifted bands make aggregate lift incomparable (Section 6.8).

Band	Baseline	Declared		Regrouped	
		Strat. $\lambda=0$	Strat.+PC $\lambda=1$	Strat. $\lambda=0$	Strat.+PC $\lambda=1$
Conservative	0.75×	0.91×	1.19×	1.21×	1.25×
Income	1.13×	1.20×	1.26×	1.25×	1.27×
Balanced	1.17×	1.22×	1.28×	1.24×	1.30×
Aggressive	1.35×	1.17×	1.97×	2.24×	2.68×

Table 6: Per-band PC-lift@10 for the five RQ3 configurations. Values above $1.00\times$ beat the random baseline $\pi(b)$.

model were the binding problem.

6.9 Limitations

- **Single backbone and operating point.** This intervention extends only LightGCN, leaving sequential, transformer, and content-based recommenders as candidates for the same treatment, and reports a single Pareto point ($\lambda = 1$) on the inherited RQ2 best-trial hyperparameters rather than re-tuning end-to-end. The reported coherence gains are therefore a lower bound on what a full frontier sweep could achieve, and the nDCG and Recall costs an upper bound.
- **Regrouping leakage and tautology.** The revealed bands are derived from each customer’s full Buy history, so they carry look-ahead information relative to each split (a per-split variant using training-window Buys only is left as future work). Because $\text{PC@}k$ is then scored against bands chosen to match those purchases, the regrouped PC@10 gain is partly guaranteed by construction. The trustworthy signal is the simultaneous nDCG@10, Recall@10, and ROI@10 gains, none of which is defined relative to the band.

7 Future Work

Beyond the per-RQ caveats noted above, several cross-cutting directions would strengthen and extend the framework.

nDCG@10 and -2.9 pp Recall@10. Margin and regrouping are partial substitutes, but only regrouping pays no ranking penalty.

We therefore take regrouped $\lambda = 0$ as the recommended operating point: it captures the bulk of the coherence gain at no ranking or return cost. Within the regrouped assignment every band also clears its own random baseline (Table 6, where regrouped $\lambda = 1$ reaches $1.25\times$ Conservative, $1.27\times$ Income, $1.30\times$ Balanced, and $2.68\times$ Aggressive).

Taken together, these results show that correcting each customer’s band (regrouping) is a cheaper and stronger lever than penalising incoherence during training (the PC-loss): once the bands are corrected there is no need to add the PC-loss signal during training, and because nDCG, Recall, and ROI rise alongside coherence, the declared bands rather than the

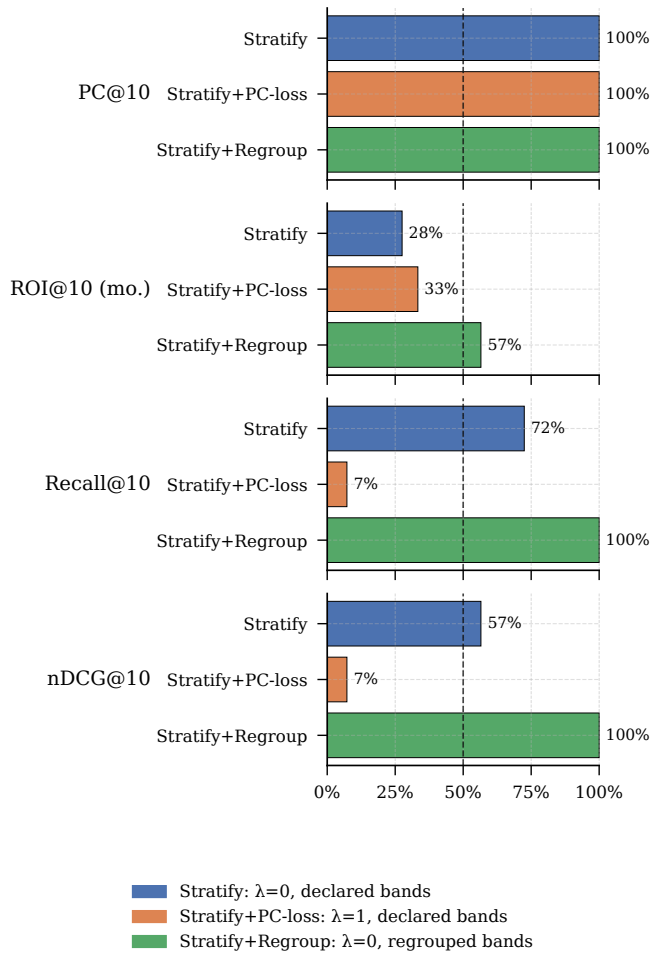


Figure 6: Per-split win rates against the global LightGCN baseline for three stratified configurations: declared $\lambda=0$, declared $\lambda=1$, and regrouped $\lambda=0$.

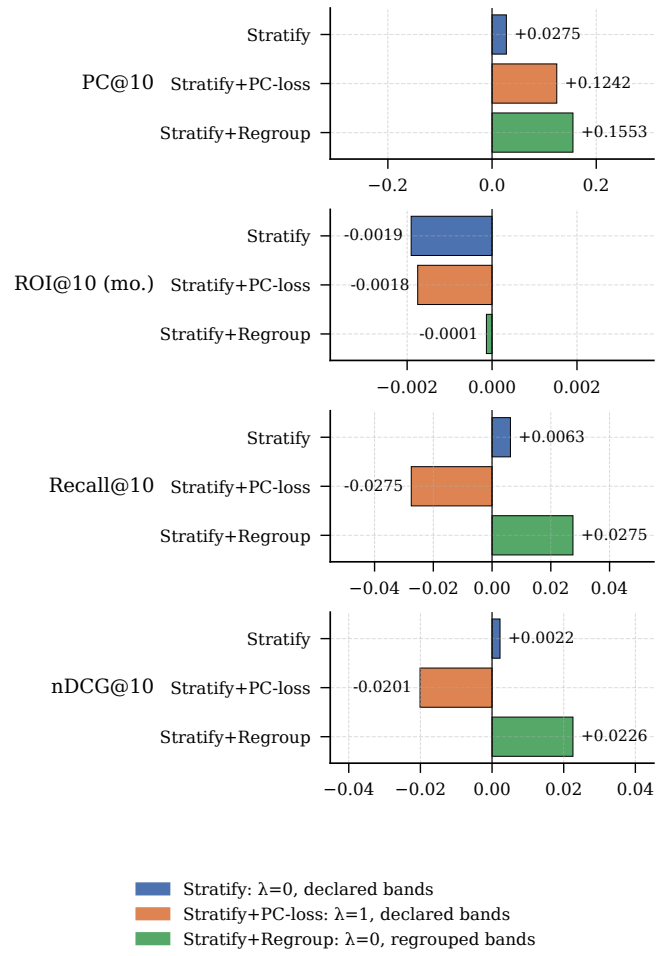


Figure 7: Paired per-split deltas against the baseline LightGCN for three stratified configurations: declared $\lambda=0$, declared $\lambda=1$, and regrouped $\lambda=0$. PC@10 is positive on every one of the 69 splits for all three.

- *Causal identification.* The RQ1 coherence premium is associational. A natural next step would be to apply causal inference techniques such as instrumental variables or difference-in-differences on datasets that span a regulatory boundary, which would also let the elevated 2018 mean discordance be tested directly against a pre-MiFID II counterfactual rather than left as a hypothesised transition effect.
- *Data-driven metric calibration.* The tolerance $d \leq 1$ and the hierarchical asset-band assignment are set rather than derived. Future work could explore data-driven tolerance calibration and validate the band assignment against external MiFID II classifications from fund providers.
- *Broader interventions and a full frontier.* The stratification and coherence-margin treatment could be applied to sequential, transformer, and content-based recommenders, and the single $\lambda = 1$ operating point could be expanded into a full Pareto sweep with end-to-end re-tuning.
- *Generalisation and online evaluation.* Results are demon-

strated on FAR-Trans under MiFID II's 4-band scale. Extending to other jurisdictions, longer horizons, or alternative asset universes would strengthen the generalisability of PC@ k . Additionally, RQ1 measures realised return on executed Buys while RQ3 measures forward return on top- k recommended slates that may never be acted on. Bridging this gap through online evaluation or counterfactual off-policy estimation is a promising direction.

8 Conclusion

The central finding of this paper is that the regulatory objective (MiFID II-aligned band coherence) and the economic objective (realised return) are not in structural conflict on FAR-Trans: a recommender can be made meaningfully more profile-coherent while keeping the return story the data already supports. This reframes coherence as an architectural choice rather than a tradeoff to be managed post-hoc, treating compliance as a training-time objective expressible as an ordinal distance on a customer-asset metric pair.

The three research questions assemble a diagnostic-to-remedy chain that supports this conclusion:

- **RQ1** establishes that the diagnostic axis is real and structural at the customer level rather than transaction-level noise, and removes the economic objection to optimising for coherence: coherent Buys earn a statistically significant return premium over discordant ones, providing evidence against the hypothesis that regulatory alignment imposes a return tax.
- **RQ2** locates the existing FAR-Trans baselines on the coherence axis and shows that both under-serve specific declared bands in ways the aggregate metrics hide.
- **RQ3** demonstrates that a profile-coherent LightGCN extension closes the per-band coverage deficit at no aggregate ROI cost. Decomposing its three components shows that regrouping customers onto their revealed band is the dominant and cost-free lever, lifting raw PC@10 to 0.949 while *also* improving nDCG and Recall, the first intervention to raise coherence and ranking together. The coherence-margin loss adds little once the bands are corrected, making stratification with regrouping the recommended configuration.

More broadly, the approach illustrates that regulatory suitability constraints can be encoded as training-time objectives rather than handled through post-hoc filtering, opening a path for recommender systems in other regulated domains (e.g., insurance product suitability, credit risk appetite matching) to embed compliance directly into their optimisation loop.

9 Reproducibility

All code, configuration grids, and evaluation artefacts are available in the project repository at <https://github.com/Samthesimpsons/SMU-Capstone>.

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